WHZ Westsächsische Hochschule Zwickau
Hochschule für Mobilität

Lab Experiment Final Report

PTI52400 ENVIRONMENTAL AND PROCESS MONITORING

Submitted To:

Prof.dr.Philip Kitschke

Engineer Nicola Pausch

Final Report of Municipal Wastewater Treatment

Submitted By Group 6

Ahmed Mustafa	56990
Sandeep Gopathoti	58590
Muhammad Wajahad	56980

Contents

1 Experimental Objectives **1**

2 Comprehensive Theoretical Framework for Wastewater Nitrogen Monitoring . . **1**

2.1 The Criticality of Nitrogen Monitoring in Environmental Engineering 1

2.1.1 Environmental Context: Eutrophication and Regulatory Drivers 2

2.2 Theoretical Fundamentals of Municipal Wastewater Treatment 2

2.2.1 The Composition of Municipal Influent 3

2.2.2 Stage 1: Mechanical Treatment (Physical Separation) 3

2.2.3 Stage 2: Biological Treatment (The Activated Sludge Process) 4

2.2.4 Stage 3: Sludge Treatment and Energy Recovery 5

2.3 Analytical Chemistry: Principles and Instrumentation 6

2.3.1 Total Nitrogen Bound (TNb) via Catalytic High-Temperature Combustion . 6

2.3.2 UV-VIS Spectroscopy and Photometry 7

3 Experimental Procedures **9**

3.1 Important Safety Reminder Before Starting 9

3.2 Determination of Total Nitrogen in Wastewater Samples 9

3.2.1 Setting Up the Instrument 9

3.2.2 Preparing Your Samples 9

3.3 Photometric Determination of Nitrite and Nitrate 10

3.3.1 Special Note on Cleanliness 10

3.3.2 Sample Preparation for Nitrate Measurement 10

3.3.3 Operating the Spectrometer: Creating a Calibration 10

3.3.4 Recording Spectra (Optional Advanced Step) 11

4 Measured Values and Results **12**

4.1 UV-VIS Spectroscopic Analysis of Nitrite Standards 12

4.2 Maximum Absorbance Values for Nitrite Standards 13

4.3 Graphical Representation of Nitrite Spectra 14

4.4 Interpretation of Spectroscopic Data 14

4.5 UV-VIS Spectroscopic Analysis of Nitrate Standards 14

4.6 Maximum Absorbance Values for Nitrate Standards 15

4.7 Graphical Representation of Nitrate Spectra 16

4.8 Interpretation of Nitrate Spectroscopic Data 17

4.9 Analysis of Mixed Nitrogen Species Sample 18

6 Mathematical Analysis of Mixed Nitrogen Species **26**
6.1 Overview of the Dual-Component Spectroscopic Challenge 26
6.2 Step 1: Extraction of Molar Absorptivity Coefficients 26
6.2.1 of Analytical Wavelengths 26
6.2.2 Linear Regression on Standard Solutions 27
6.3 Step 2: Measurement of the Mixed Sample 27
6.4 Step 3: Formulation of Simultaneous Equations 27
6.5 Step 4: Algebraic Solution Using Cramer's Rule 28
6.5.1 Calculation of the Determinant 28
6.5.2 Solution for Nitrate Concentration 28
6.5.3 Solution for Nitrite Concentration 28
6.6 Step 5: Multi-Wavelength Least Squares Refinement 29
6.6.1 Mathematical Formulation 29
6.6.2 Numerical Implementation 29
6.6.3 Optimized Results 30
6.6.4 Statistical Validation 30
6.7 Summary of Analytical Results 30
6.8 Physical Interpretation and Validation 30
6.8.1 Mass Concentration Determination 31
6.8.2 Nitrogen Removal Efficiency 31
6.8.3 Conversion to Nitrogen-Equivalent Concentrations 31
6.8.4 Nitrogen Balance Summary 32

7 Discussion and Results **32**
7.1 Evaluation of Analytical Method Performance 32
7.2 Selectivity Assessment 32
7.2.1 Spectral Overlap in UV Spectroscopy 32
7.2.2 Selectivity in Photometric Analysis 33
7.3 Sensitivity Comparison 34
7.3.1 Calibration Slopes from Experimental Data 34

8 Conclusion **34**

List of Figures

1 UV-VIS absorption spectra of nitrite standard solutions showing absorbance as a function of wavelength for concentrations of 0.25, 1.00, 2.00, and 3.00 mg L^{-1} . 14
2 Maximum Absorbance Values for Each Nitrite Concentration 14
3 Calibration for NO2 using photometric analysis 15
4 UV-VIS absorption spectra of nitrate standard solutions showing absorbance as a function of wavelength for concentrations of 0.25, 1.00, 2.00, and 3.00 mg L^{-1} . 17
5 Maximum absorbance values for each nitrate concentration 17
6 Calibration curve for $NO3^-$ using photometric analysis 17
7 UV-VIS absorption spectrum of mixed nitrogen species sample showing combined $NO2^-$ and $NO3^-$ absorbance as a function of wavelength 18
8 UV absorption spectrum of Sample 2 (mixed NO_2^- and NO_3^-) showing a single overlapping peak that prevents direct differentiation between the two nitrogen species 24

List of Tables

1 Comparison of Key Biological Nitrogen Removal Processes 4
2 UV-VIS Absorption Spectra of Nitrite Standard Solutions (Part 1: 190-245 nm) . 12
3 UV-VIS Absorption Spectra of Nitrite Standard Solutions (Part 2: 250-300 nm) . 13
4 Maximum Absorbance Values for Each Nitrite Concentration 13
5 UV-VIS Absorption Spectra of Nitrate Standard Solutions (190-245 nm) 15
6 UV-VIS Absorption Spectra of Nitrate Standard Solutions (250-300 nm) 16
7 Maximum Absorbance Values for Each Nitrate Concentration 16
8 Comparison of Two-Wavelength and Multi-Wavelength Analysis Results 23
9 Comparison of Analytical Sensitivity Between UV Spectroscopy and Photometric Analysis Methods 26

1 Experimental Objectives

The primary goal of this practical training was to evaluate analytical methods for nitrogen monitoring in wastewater. The specific experimental tasks were defined as follows:

1. **Calibration and Spectral Analysis:** Establish calibration functions for nitrite and nitrate using standard solutions ($w = 0.25 - 3.00$ mg L^{-1}). Record UV absorption spectra ($\lambda = 190 - 300$ nm) to determine absorbance maxima.
2. **Method Comparison:** Analyze a binary mixture containing 1.00 mg L^{-1} of both nitrite and nitrate. Compare the selectivity and sensitivity of the photometric method (Griess reaction) against direct UV spectroscopy.
3. **Wastewater Analysis:** Determine Total Nitrogen (TNb), nitrate, and nitrite concentrations in samples from the inlet, aeration tank, and outlet of a municipal treatment plant.
   Note: Analysis of wastewater samples could not be completed due to insufficient data.
4. **Limit of Detection (LOD):** Perform a 10-fold determination of a low-concentration standard ($w = 0.10$ mg L^{-1}) to estimate the detection limit.
   Note: This statistical determination was not performed due to missing experimental data.

2 Comprehensive Theoretical Framework for Wastewater Nitrogen Monitoring

2.1 The Criticality of Nitrogen Monitoring in Environmental Engineering

The effective management of municipal wastewater represents one of the defining challenges of modern civil and environmental engineering [1]. As urbanization accelerates and population densities increase, the anthropogenic load on the hydrological cycle intensifies, necessitating robust, efficient, and scientifically rigorous treatment protocols. The "Practical Training: Environmental and Process Monitoring" course, specifically Experiment I focusing on "Monitoring parameters of a municipal wastewater treatment plant," serves as a microcosm of this broader engineering challenge. It bridges the theoretical understanding of biochemical transformation processes with the practical rigors of analytical chemistry and occupational safety.

This report provides an exhaustive analysis of the theoretical frameworks, safety architectures, and documentation standards required for this practical course. It is designed not merely as a guide but as a foundational reference document that synthesizes the complex interplay between biological wastewater treatment, photometric analysis, and rigorous safety compliance.

2.1.1 Environmental Context: Eutrophication and Regulatory Drivers

To understand the necessity of the monitoring parameters discussed—specifically Total Nitrogen (TNb), Nitrate (NO_3^-), and Nitrite (NO_2^-)—one must first contextualize the environmental impact of untreated effluent [2]. Municipal wastewater is a potent cocktail of nutrients, primarily nitrogen and phosphorus. Nitrogen enters the waste stream largely through human metabolic waste (urea) and protein degradation.

When discharged into surface waters, these nutrients disrupt the oligotrophic equilibrium of aquatic ecosystems. Nitrogen acts as a limiting nutrient; its sudden abundance triggers eutrophication, a phenomenon characterized by explosive algal blooms [1]. The consequences are cascading and catastrophic:

- **Algal Proliferation:** Rapid growth of phytoplankton blocks sunlight, inhibiting the photosynthesis of submerged macrophytes.
- **Oxygen Depletion:** As the short-lived algae die, they sink to the benthic zone where heterotrophic bacteria decompose the biomass. This decomposition is an oxygen-consuming process.
- **Hypoxia and Anoxia:** The bacterial consumption of dissolved oxygen (DO) creates hypoxic "dead zones" where aquatic fauna, including fish and benthic invertebrates, cannot survive.

Recognizing this threat, the European Union established Council Directive 91/271/EEC concerning urban wastewater treatment [3]. This directive mandates rigorous effluent standards, particularly for "sensitive areas" susceptible to eutrophication. For large treatment plants (serving a population equivalent of over 100,000), the directive imposes a strict limit of 10 mg/L for total nitrogen. Compliance with these regulations is not optional; it is a legal imperative that drives the design and operation of every municipal treatment plant in the EU. Consequently, the ability to accurately monitor nitrogen species is not just an academic exercise for students; it is a fundamental skill for ensuring legal compliance and environmental stewardship.

2.2 Theoretical Fundamentals of Municipal Wastewater Treatment

The wastewater treatment plant (WWTP) is essentially a managed ecosystem where engineering controls are applied to accelerate natural self-purification processes [2]. The theoretical section of the student protocol must distill complex biochemical kinetics into clear, understandable concepts.

2.2.1 The Composition of Municipal Influent

Municipal wastewater is a heterogeneous mixture. While it is 99.9% water, the remaining 0.1% defines the treatment challenge [1]:

- **Domestic Effluent:** Contains feces, urine, food scraps, and detergents. This fraction contributes the bulk of the organic carbon (measured as BOD/COD) and nutrients (nitrogen and phosphorus).
- **Rainwater and Runoff:** Introduces grit, sand, heavy metals from road wear, and hydrocarbons.
- **Infiltration Water:** Groundwater leaking into aging sewer pipes can dilute the influent.

The monitoring documents highlight that the influent contains a "plethora of nutrients," including carbon compounds and urea (CH_4N_2O) [4]. Urea is rapidly hydrolyzed by the enzyme urease (ubiquitous in sewer systems) into ammonium and carbonate:

$$\mathrm{CO(NH_2)_2 + 2\,H_2O \xrightarrow{\text{urease}} 2\,NH_4{}^+ + CO_3{}^{2-}} \tag{1}$$

This means that by the time wastewater reaches the plant, most nitrogen is already in the form of ammonium ($NH_4{}^+$).

2.2.2 Stage 1: Mechanical Treatment (Physical Separation)

The first line of defense is purely physical. This stage protects the downstream biological assets from damage and clogging [2]:

Screening Systems: Coarse and fine screens intercept rags, plastics, and hygiene products. These materials are non-biodegradable and would wreak havoc on pumps and aerators.

The Aerated Grit and Grease Chamber: This is a crucial unit operation where fluid dynamics are manipulated to separate three phases. By reducing the flow velocity to roughly 0.3 m/s, the plant achieves a specific settling regime:

- Heavy inorganic solids (sand, gravel, and glass shards) settle to the hopper bottom.
- Light organic liquids (fats, oils, and greases, FOG) float to the surface due to their lower density ($\rho < 1$ g/cm).
- Suspended organics (feces and food particles) remain suspended and pass to the next stage.

The "Monitoring parameters" document explicitly notes that fat removal is critical because FOG can coat biological flocs, hindering oxygen transfer in the aeration tank [4].

2.2.3 Stage 2: Biological Treatment (The Activated Sludge Process)

This stage is the metabolic engine of the plant. It utilizes a suspended growth of microorganisms—the "activated sludge"—to mineralize dissolved pollutants. The process relies on a complex microbial consortium comprising bacteria, protozoa, and rotifers [1].

Table 1: Comparison of Key Biological Nitrogen Removal Processes

Parameter	Nitrification	Denitrification
Microorganisms	*Nitrosomonas*, *Nitrobacter* (autotrophs)	*Pseudomonas*, *Paracoccus* (heterotrophs)
Oxygen Requirement	Aerobic (DO > 2 mg/L)	Anoxic (DO < 0.2 mg/L)
Carbon Source	Inorganic (CO_2)	Organic (BOD)
pH Effect	Consumes alkalinity, lowers pH	Produces alkalinity, raises pH
Optimum Temperature	25-30C	20-30C
Energy Yield	Low	High

Heterotrophic Carbon Oxidation: The majority of the microbial biomass consists of heterotrophic bacteria. These organisms require organic carbon as both an energy source and a carbon source for cell synthesis. In the presence of dissolved oxygen (supplied via aeration), they oxidize organic matter:

$$\text{Substrate } (\text{C}_{org}) + \text{O}_2 + \text{Nutrients} \xrightarrow{\text{Heterotrophs}} \text{CO}_2 + \text{H}_2\text{O} + \text{Biomass (Sludge)} + \text{Energy} \quad (2)$$

This reaction reduces the Biochemical Oxygen Demand (BOD) of the water. The document emphasizes that this is a non-specific process carried out by a wide range of aerobic organisms [4].

Nitrogen Elimination: The Two-Step Biological Dance: Nitrogen removal is more distinct and sensitive than carbon removal. It involves two sequential biological processes that require diametrically opposed environmental conditions: Nitrification (aerobic) and Denitrification (anoxic).

A. Nitrification (Aerobic Autotrophic Oxidation): Nitrification is performed by autotrophic bacteria, which means they use inorganic carbon (CO_2) for cell growth and derive energy from the oxidation of reduced nitrogen compounds. This process is highly sensitive to environmental stressors (pH, temperature, toxicity).

Ammonia Oxidation: Bacteria of the genus *Nitrosomonas* oxidize ammonium (NH_4^+) to nitrite

(NO_2^-):

$$2\,NH_4^+ + 3\,O_2 \longrightarrow 2\,NO_2^- + 2\,H_2O + 4\,H^+ + \text{Energy} \tag{3}$$

Note: The production of hydrogen ions (H^+) consumes alkalinity, lowering the pH of the mixed liquor. For every mole of ammonium oxidized, two moles of acidity are produced. If the wastewater has low buffering capacity, the pH can drop, inhibiting the nitrifiers themselves [1].

Nitrite Oxidation: Bacteria of the genus *Nitrobacter* (and *Nitrospira*) oxidize nitrite to nitrate (NO_3^-):

$$2\,NO_2^- + O_2 \longrightarrow 2\,NO_3^- + \text{Energy} \tag{4}$$

Note: This step yields very little energy, causing *Nitrobacter* to grow slowly. It is critical that this step keeps pace with the first; otherwise, toxic nitrite accumulates.

B. Denitrification (Anoxic Heterotrophic Reduction): Nitrification only converts the nitrogen form (from toxic ammonia to less toxic nitrate); it does not remove nitrogen from the water. Removal requires converting soluble nitrate into inert nitrogen gas (N_2), which escapes into the atmosphere.

This process is performed by facultative heterotrophs (e.g., *Pseudomonas*, *Paracoccus*) under anoxic conditions. "Anoxic" means dissolved oxygen is absent (DO < 0.2 mg/L), but chemically bound oxygen (in nitrate) is available. The bacteria preferentially use dissolved oxygen if available; only when DO is depleted do they switch their metabolic pathway to use nitrate as the terminal electron acceptor [2].

$$4\,NO_3^- + 4\,H^+ + 5\,C_{org} \longrightarrow 2\,N_2\uparrow + 5\,CO_2 + 2\,H_2O \tag{5}$$

Critical Insight: The reaction equation highlights the absolute necessity of a carbon source (C_{org}). In the "Monitoring parameters" document, it is noted that denitrification typically happens upstream (pre-anoxic zone) where raw wastewater provides the carbon, or via intermittent aeration where the biomass uses stored internal carbon. If carbon is insufficient (low C:N ratio), denitrification fails, and nitrate is discharged. External carbon sources like methanol or acetate may be added in such cases [4].

2.2.4 Stage 3: Sludge Treatment and Energy Recovery

The biological process produces excess biomass (waste activated sludge). This sludge is chemically unstable and pathogenic [2].

Anaerobic Digestion: The sludge is pumped to digestion towers (bioreactors) operating in the absence of oxygen. Here, anaerobic archaea break down the organic cell walls, converting them into biogas (roughly 60-65% methane, 35-40% CO_2).

Energy Recovery: This methane is a valuable renewable fuel. It is typically burned in Combined Heat and Power (CHP) units to generate electricity and heat for the plant, often making the facility energy-neutral.

Disposal: The remaining "digested" sludge is stabilized (reduced pathogens and odors). It is dewatered using centrifuges or belt presses and then incinerated or used as a soil conditioner (biosolids), depending on heavy metal content regulations.

2.3 Analytical Chemistry: Principles and Instrumentation

The core of the student's experimental work involves the quantification of nitrogen species. The protocol requires a deep understanding of the chemical and physical principles underlying these measurements.

2.3.1 Total Nitrogen Bound (TNb) via Catalytic High-Temperature Combustion

The determination of TNb represents a significant advancement over the historical Kjeldahl method [5]. The Kjeldahl method only measures organic nitrogen and ammonium (TKN) and requires boiling sulfuric acid with metallic catalysts for hours. The catalytic combustion method used in this course is rapid, covers all nitrogen species (including nitrate/nitrite), and minimizes hazardous waste.

The Combustion Process: The wastewater sample is injected directly into a furnace heated to approximately 800C. The carrier gas is oxygen.

Catalysis: A platinum-based catalyst ensures the complete oxidation of all nitrogen forms. Even refractory organic nitrogen (like pyridine rings) is broken down.

Reaction:

$$R - N + \mathrm{O}_2 \xrightarrow{Pt, \Delta T} \mathrm{CO}_2 + \mathrm{H}_2\mathrm{O} + \mathrm{NO} + \text{oxides} \tag{6}$$

The key outcome is that effectively all nitrogen is converted into Nitrogen Monoxide (NO).

Gas Conditioning: Before detection, the gas stream must be conditioned:

- **Water Removal:** A Peltier cooler condenses water vapor, which would otherwise interfere with the detector.
- **Halogen Scrubbing:** A copper/brass trap removes chlorine and other halogens that could corrode the sensor.

Chemiluminescence Detection (CLD): This is the physical heart of the analysis. The principle relies on the specific reaction between nitric oxide and ozone.

Ozone Generation: An internal generator produces ozone (O_3) from the oxygen carrier gas.

Reaction: NO from the sample meets O_3 in the reaction chamber:

$$\mathrm{NO} + \mathrm{O_3} \longrightarrow \mathrm{NO_2}^* + \mathrm{O_2} \tag{7}$$

The asterisk (*) denotes that the Nitrogen Dioxide produced is in an electronically excited (high energy) state.

Light Emission: This excited state is unstable. The molecule rapidly relaxes to its ground state by emitting a photon:

$$\mathrm{NO_2}^* \longrightarrow \mathrm{NO_2} + h\nu\,(\text{light}) \tag{8}$$

The wavelength of this light is in the near-infrared/visible range (600-3000 nm).

Quantification: A photomultiplier tube (PMT) detects these photons. The intensity of the light is strictly proportional to the number of NO molecules, and thus to the original nitrogen concentration in the sample. This linear dynamic range allows for precise calibration.

2.3.2 UV-VIS Spectroscopy and Photometry

For the differentiation of nitrogen species (Nitrate vs. Nitrite), the course employs UV-VIS photometry. This relies on the interaction between light and the electron clouds of molecules [5].

The Physics of Absorption: When light passes through a solution, photons with specific energy levels can be absorbed by electrons in the solute, promoting them to higher energy orbitals ($\pi \to \pi^*$ or $n \to \pi^*$ transitions). The "Monitoring parameters" document notes that nitrate and nitrite have intrinsic absorption in the UV range (around 210-220 nm), but this region is prone to interference from organic matter (which also absorbs in the UV). Therefore, colorimetric derivatization in the visible range is preferred for wastewater.

The relationship between absorption and concentration is governed by the Lambert-Beer Law:

$$E = \log\left(\frac{I_0}{I_T}\right) = \varepsilon \cdot c \cdot d \tag{9}$$

- E: Extinction (Absorbance), a dimensionless quantity
- I_0: Intensity of incident light
- I_T: Intensity of transmitted light
- ε: Molar extinction coefficient (L·mol^{-1}·cm^{-1}), a constant specific to the molecule and wavelength
- c: Concentration (mol/L)

- d: Path length of the cuvette (typically 1 cm)

Nitrite Determination: The Griess Reaction: Nitrite (NO_2^-) is determined via a specific diazotization reaction, known historically as the Griess test. This reaction is highly selective and sensitive [5].

Step 1: Diazotization. In an acidic environment (pH < 2), nitrite becomes nitrous acid (HNO_2). This reacts with a primary aromatic amine—in this protocol, Sulfanilamide. The nitrite attacks the amine group, eliminating water and forming a diazonium ion:

$$\text{Sulfanilamide} + \mathrm{NO_2}^- + 2\mathrm{H}^+ \rightarrow \text{Diazonium Salt} + 2\mathrm{H_2O} \tag{10}$$

Step 2: Azo Coupling. The diazonium salt is an electrophile. It attacks the electron-rich aromatic ring of the coupling agent, N-(1-naphthyl)-ethylenediamine dihydrochloride (NED). This coupling forms an azo bond (-N=N-) bridging the two rings:

$$\text{Diazonium Salt} + \text{NED} \rightarrow \text{Red-Violet Azo Dye} + \mathrm{H}^+ \tag{11}$$

Detection: This resulting azo dye has an extended conjugated π-electron system, which shifts its absorption maximum into the visible green region ($\lambda \approx 540$ nm). The solution appears red-violet to the eye. The intensity of this color is directly proportional to the nitrite concentration.

Nitrate Determination: Reduction to Nitrite: Nitrate (NO_3^-) is chemically inert to the Griess reagents. To measure it, it must first be reduced to nitrite.

The Reductant: The protocol uses Hydrazine ($H_2N{-}NH_2$) as the reducing agent.

Conditions: The reaction requires an alkaline environment (pH $\approx$ 10-11) and the presence of Copper (Cu^{2+}) and Zinc (Zn^{2+}) ions as catalysts.

Stoichiometry:

$$2\,\mathrm{NO_3}^- + \mathrm{H_2N{-}NH_2} \xrightarrow{\text{Cu/Zn}} 2\,\mathrm{NO_2}^- + \mathrm{N_2} + 2\,\mathrm{H_2O} \tag{12}$$

Calculation: The analytical process involves two parallel measurements:

1. Measure Nitrite directly (without reduction)
2. Measure Total Oxidized Nitrogen (Nitrate + Nitrite) after reduction
3. Nitrate $=$ Total Oxidized Nitrogen $-$ Original Nitrite

Interferences: The document highlights that organic matter (giving a brown color) and turbidity (suspended solids) interfere. Filtration (0.45 m) is mandatory to remove solids. "Background correction" can be done by measuring the sample without reagents to subtract its intrinsic color.

3 Experimental Procedures

3.1 Important Safety Reminder Before Starting

Before you begin any part of this experiment, you must wear your protective equipment. This means safety goggles, gloves (nitrile or butyl rubber), and a lab coat. These are not optional—they are essential because we work with dangerous chemicals like hydrazine. Remember the safety rules from the previous section and think carefully before each step.

3.2 Determination of Total Nitrogen in Wastewater Samples

This part uses the combustion analyzer to measure all forms of nitrogen in your samples.

3.2.1 Setting Up the Instrument

1. **Select the Method:** Open the instrument software. Go to the "Method" tab and select "Open Method." Choose the file named "NP_TN_2024-11."
2. **Start a New Measurement:** Click on the "Measurement" tab, then select "Start Measurement." The computer will ask you to name your analysis table—use today's date (for example: 2024-11-15). First, measure the preparation water (blank) to make sure the instrument is clean and ready.
3. **Prepare Your Sample List:** Go to the "Rack Table" tab and click "New." Name your samples clearly so you know which is which: "Inlet," "Aeration Basin," and "Outlet."

3.2.2 Preparing Your Samples

1. **Filter the Samples:** You must remove all solid particles from your wastewater samples. Use a syringe filter with a pore size of 0.45 micrometers (made of polyether sulfone). Push each sample through the filter into a clean vial.
2. **Load the Autosampler:** Place your filtered samples into the autosampler tray according to the positions you specified in your rack table.
3. **Start the Analysis:** Confirm the measurement order by clicking "OK." The instrument will now automatically inject each sample into the hot furnace, burn it, and measure the nitrogen content.

3.3 Photometric Determination of Nitrite and Nitrate

This is the colorimetric method where we turn nitrate and nitrite into a pink color and measure how strong that color is.

3.3.1 Special Note on Cleanliness

Nitrate determination is very sensitive to contamination. Even tiny amounts of dirt can ruin your results. Before you start, rinse all your glassware thoroughly with deionized water. This includes test tubes, pipettes, and cuvettes.

3.3.2 Sample Preparation for Nitrate Measurement

For each sample and standard solution, you need to follow these steps exactly:

1. **Start with 10 mL:** Place exactly 10 mL of your sample or standard into a clean test tube.
2. **Add the Catalysts:** Using a pipette, add 1 mL of the catalyst solution. Gently mix the tube by turning it upside down once. Then add 1 mL of sodium hydroxide solution and mix again. Finally, add 1 mL of the hydrazine solution.
3. **Mix and Wait:** Now shake the test tube vigorously for a few seconds. Set a timer for exactly 5 minutes. When the timer goes off, shake the tube again.
4. **Add the Color-Forming Reagents:** After exactly 5 minutes, add 2 mL of sulfanilamide solution and 1 mL of NED solution. Mix well.
5. **Wait for Color Development:** Let the mixture sit for 3 minutes. During this time, the solution will turn red-violet if nitrate (or nitrite) is present. The color intensity tells us the concentration.
6. **Measure the Color:** After 3 minutes, transfer the solution to a cuvette and measure its absorbance in the spectrometer at a wavelength of 541 nm.

Important Note for Nitrite Only: If you only want to measure nitrite (not nitrate), you can skip the first steps with the catalysts and hydrazine. For nitrite alone, simply add the sulfanilamide (2 mL) and NED (1 mL) directly to your 10 mL sample, wait 3 minutes, and measure.

3.3.3 Operating the Spectrometer: Creating a Calibration

To convert your absorbance readings into concentrations, you need to create a calibration curve using known standard solutions.

1. **Set Up the Measurement:** On the spectrometer computer, go to the "Measurement" tab and select the parameter set called "2024-11-04-NP-Nitrat."
2. **Start the Calibration:** Go to the "Quant" tab and click "Calibration." You need to set the following parameters:
   - Regression model: Choose $y = A + B \cdot x$ (this is a straight line)
   - Wavelength: Set to 541 nm
   - Unit: Choose mg/L (which is the same as ppm)
   - Cuvette thickness: 1 cm
   - Number of standards: 5
3. **Enter Standard Concentrations:** Click the "Standards" button and type in the concentrations of your five standard solutions (these will be given to you).
4. **Prepare the Cuvette:** Use a quartz glass cuvette. Fill it about three-quarters full with deionized water, swirl it around to rinse, and pour the water out. Do this twice. Then fill the cuvette with fresh deionized water for your blank measurement. Make sure there are no air bubbles.
5. **Handle the Cuvette Properly:** Wipe the outside of the cuvette with a clean microfiber cloth. Only touch the matte sides, not the clear sides where light passes through. Place it in the spectrometer.
6. **Measure the Blank:** Press the "Reference" button. The machine will measure the blank three times to set the baseline.
7. **Measure the Standards:** Press "Start." The machine will prompt you to put your first standard into the cuvette. Empty the cuvette, rinse it with a little of your standard solution, then fill it with the standard. Place it in the spectrometer and click "OK." Repeat this for all five standards.
8. **Check Your Calibration:** The computer will draw a calibration line. A good calibration should be a straight line. If it looks good, you can now measure your samples.
9. **Measure Your Samples:** Follow the same procedure with your inlet, aeration basin, and outlet samples. The computer will use your calibration line to calculate their concentrations automatically.

Repeat for Nitrite: You need to do the exact same calibration procedure for nitrite. Just select the nitrite parameter set instead of the nitrate one.

3.3.4 Recording Spectra (Optional Advanced Step)

You can also record full spectra of your solutions to see how they absorb light at different wavelengths.

1. **Select the Spectrum Mode:** Go to the "Measurement" tab and choose the parameter set "2024-11-04-NP-Nitrat-Spek.par."

2. **Record a Reference Spectrum:** Fill a clean cuvette with deionized water (following the same rinsing procedure as before). Place it in the spectrometer and record a reference spectrum.

3. **Record Sample Spectra:** Now replace the water with your standard solutions, your samples, or even the colored solutions from your nitrate/nitrite measurements. Record and save each spectrum.

4. **Save Your Data:** Go to "File" "Export" and save your measured values as a CSV file. This lets you open the data in Excel or other programs for further analysis.

4 Measured Values and Results

4.1 UV-VIS Spectroscopic Analysis of Nitrite Standards

The photometric determination of nitrite was carried out using the Griess reaction (Section 3.2). Four standard solutions (0.25, 1.00, 2.00, and 3.00 mg/L) were analyzed to evaluate the relationship between nitrite concentration and absorbance in accordance with the Lambert–Beer Law.

Absorbance spectra were recorded at 1 nm intervals over the selected wavelength range to examine spectral characteristics and potential interferences. The complete spectroscopic data for all standards are presented in Table 3.

Table 2: UV-VIS Absorption Spectra of Nitrite Standard Solutions (Part 1: 190-245 nm)

Wavelength / nm	**0.25 mg L^{-1}**	**1.00 mg L^{-1}**	**2.00 mg L^{-1}**	**3.00 mg L^{-1}**
190	0.01162	0.051174	0.142349	0.207681
195	0.016947	0.046314	0.149479	0.235848
200	0.019759	0.061332	0.176582	0.281039
205	0.022879	0.078067	0.204658	0.323693
210	0.025322	0.086602	0.21473	0.34198
215	0.02288	0.081117	0.198224	0.317056
220	0.018184	0.063019	0.158119	0.254178
225	0.01358	0.038038	0.106761	0.174068
230	0.006513	0.013599	0.057261	0.09887
235	0.003096	-0.004208	0.02227	0.043538
240	-0.000322	-0.013542	0.002284	0.012401
245	-0.000303	-0.017783	-0.006295	-0.002335

Note: Tables 3 and 4 present selected wavelengths at five-nanometer intervals for clarity. The

Table 3: UV-VIS Absorption Spectra of Nitrite Standard Solutions (Part 2: 250-300 nm)

Wavelength / nm	**0.25 mg L^{-1}**	**1.00 mg L^{-1}**	**2.00 mg L^{-1}**	**3.00 mg L^{-1}**
250	-0.001356	-0.01873	-0.009112	-0.007653
255	-0.000936	-0.017918	-0.00837	-0.007394
260	-0.001684	-0.017519	-0.009052	-0.007466
265	-0.001765	-0.018069	-0.009145	-0.00633
270	-0.00057	-0.017847	-0.007131	-0.00705
275	-0.002527	-0.017658	-0.006854	-0.005054
280	-0.0034	-0.016112	-0.00721	-0.004561
285	-0.004489	-0.018627	-0.009813	-0.007864
290	-0.001946	-0.016436	-0.006722	-0.004286
295	-0.002416	-0.014583	-0.004238	-0.003806
300	-0.000829	-0.018018	-0.00983	-0.00699

complete dataset with measurements at every nanometer from 190 to 300 nm is available in the electronic supplementary material or upon request.

4.2 Maximum Absorbance Values for Nitrite Standards

The maximum absorbance values for each nitrite standard concentration were identified from the complete spectral scan. These values are critical for establishing the calibration curve and determining the optimal measurement wavelength. Table 5 presents the maximum absorbance values and their corresponding wavelengths for all four standard concentrations.

Table 4: Maximum Absorbance Values for Each Nitrite Concentration

$NO2^-$ Concentration / mg L^{-1}	**Maximum Absorbance**	**Wavelength / nm**
0.25	0.026290	209
1.00	0.088156	211
2.00	0.215772	209
3.00	0.342849	209

The data in Table 5 demonstrate that the maximum absorbance for all nitrite concentrations occurs in the narrow wavelength range of 209 to 211 nm, which is consistent with the expected UV absorption characteristics of the azo dye formed during the Griess reaction. The linear increase in maximum absorbance with increasing nitrite concentration confirms the validity of the Lambert-Beer Law for this analytical method and indicates that the method is suitable for quantitative nitrite determination across the tested concentration range.

4.3 Graphical Representation of Nitrite Spectra

This subsection contains the graphical plots of the UV-VIS absorption spectra for all nitrite standard solutions. The graphs visually illustrate the relationship between wavelength and absorbance.

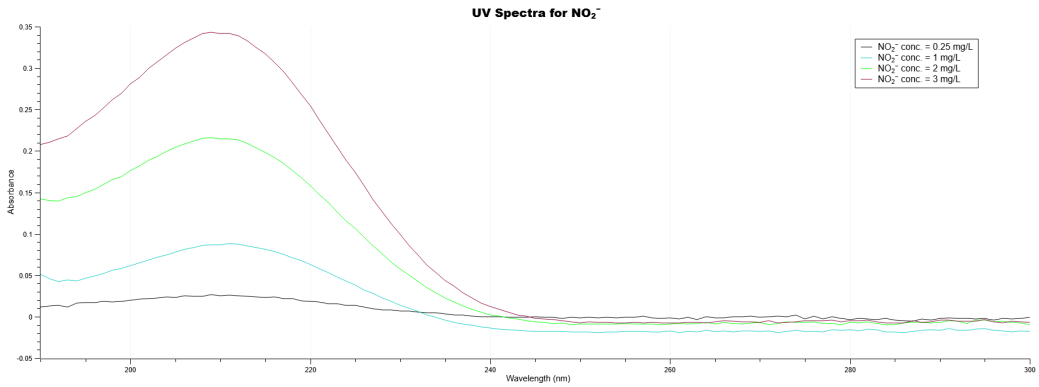

Figure 1: UV-VIS absorption spectra of nitrite standard solutions showing absorbance as a function of wavelength for concentrations of 0.25, 1.00, 2.00, and 3.00 mg L^{-1}

4.4 Interpretation of Spectroscopic Data

The spectroscopic measurements reveal several important characteristics of the nitrite determination method. In the UV region (190-230 nm), all standard solutions show positive absorbance values that increase with concentration, indicating strong absorption in this range.

The negative absorbance values observed at wavelengths above 235 nm are artifacts of the baseline correction and instrumental noise in regions where the sample shows minimal absorption.

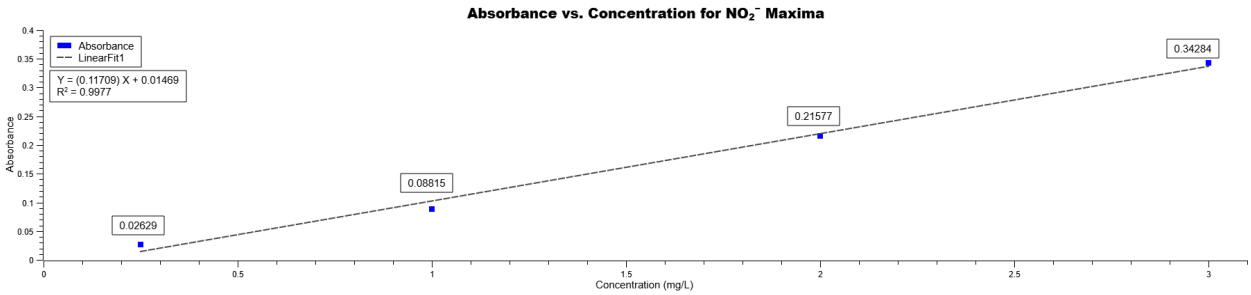

Figure 2: Maximum Absorbance Values for Each Nitrite Concentration

4.5 UV-VIS Spectroscopic Analysis of Nitrate Standards

The photometric determination of nitrate was performed using the reduction method followed by the Griess reaction as described in Section 3.2. Four standard solutions with nitrate concentrations of 0.25, 1.00, 2.00, and 3.00 mg/L were prepared and measured across the wavelength

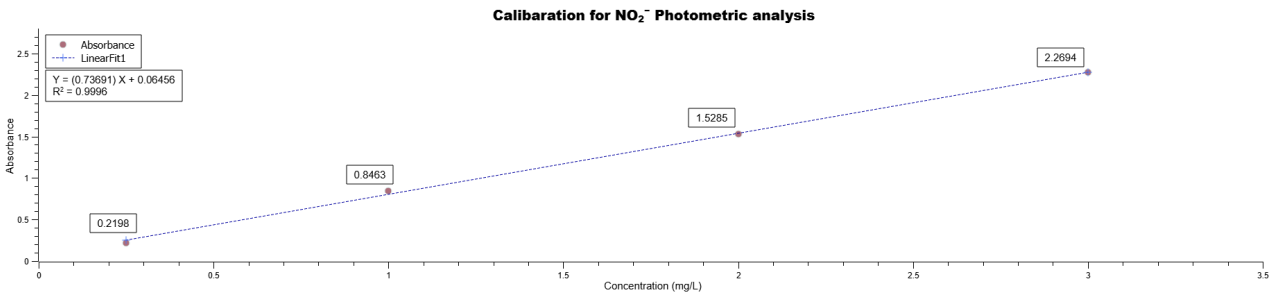

Figure 3: Calibration for NO2 using photometric analysis

range of 190 to 300 nm. The absorbance spectra were recorded to characterize the spectral behavior of the nitrate-derived azo dye and to establish the calibration parameters.

Tables 6 and 7 present the spectroscopic data for all four nitrate standard solutions at selected wavelengths. The measurements were conducted to verify the linearity of the method and to identify any potential spectral interferences.

Table 5: UV-VIS Absorption Spectra of Nitrate Standard Solutions (190-245 nm)

Wavelength / nm	**0.25 mg L^{-1}**	**1.00 mg L^{-1}**	**2.00 mg L^{-1}**	**3.00 mg L^{-1}**
190	0.030189	0.128073	0.2425	0.361006
195	0.040438	0.155821	0.297648	0.442741
200	0.040146	0.164315	0.315518	0.473783
205	0.039069	0.157154	0.305712	0.458986
210	0.031913	0.135857	0.262084	0.394764
215	0.026185	0.101207	0.196351	0.292953
220	0.016605	0.063976	0.119826	0.180171
225	0.008109	0.033584	0.060514	0.09171
230	0.001321	0.014446	0.024746	0.038263
235	-0.001658	0.005415	0.007727	0.013442
240	-0.001455	0.003054	0.002992	0.005897
245	-0.003274	0.001473	-0.000662	0.000439

Note: Tables 6 and 7 present selected wavelengths at five-nanometer intervals for clarity. The complete dataset with measurements at every nanometer from 190 to 300 nm is available in the electronic supplementary material or upon request.

4.6 Maximum Absorbance Values for Nitrate Standards

The maximum absorbance values for each nitrate standard concentration were identified from the complete spectral scan. These values are essential for establishing the calibration curve for nitrate determination. Table 8 presents the maximum absorbance values and their corresponding wavelengths for all four standard concentrations.

The data in Table 8 demonstrate that the maximum absorbance for all nitrate concentrations

Table 6: UV-VIS Absorption Spectra of Nitrate Standard Solutions (250-300 nm)

Wavelength / nm	**0.25 mg L^{-1}**	**1.00 mg L^{-1}**	**2.00 mg L^{-1}**	**3.00 mg L^{-1}**
250	-0.00384	0.000132	-0.000937	-0.001749
255	-0.003449	-0.000402	-0.001728	-0.003327
260	-0.003459	0.001225	-0.002671	-0.005758
265	-0.002535	0.003621	-0.002539	-0.005217
270	-0.002075	0.00453	-0.004337	-0.004642
275	-0.002497	0.005146	0.000249	-0.004366
280	-0.003856	0.001851	-0.003981	-0.006427
285	-0.000526	0.0023	-0.001517	-0.001655
290	-0.001671	0.0058	-0.001151	-0.005438
295	0.000216	0.00848	0.000546	-0.002923
300	0.000092	0.004751	-0.003165	-0.005012

Table 7: Maximum Absorbance Values for Each Nitrate Concentration

$NO3^-$ Concentration / mg L^{-1}	**Maximum Absorbance**	**Wavelength / nm**
0.25	0.041050	201
1.00	0.164315	200
2.00	0.316228	201
3.00	0.473783	200

occurs in the narrow wavelength range of 200 to 201 nm. This is consistent with the UV absorption characteristics of the azo dye formed after the reduction of nitrate to nitrite and subsequent Griess reaction. The linear increase in maximum absorbance with increasing nitrate concentration confirms the validity of the Lambert-Beer Law for this analytical method and indicates that the nitrate determination procedure, including the reduction step, is suitable for quantitative analysis across the tested concentration range.

4.7 Graphical Representation of Nitrate Spectra

This subsection contains the graphical plots of the UV-VIS absorption spectra for all nitrate standard solutions. The graphs visually illustrate the relationship between wavelength and absorbance, and demonstrate the concentration-dependent absorption behavior of the nitrate-derived azo dye following the reduction and color formation steps.

Note: To add your graphs, uncomment the `\includegraphics` lines above and replace the filenames with your actual image files. Make sure the image files are in the same directory as your LaTeX document, or provide the correct path to the image files.

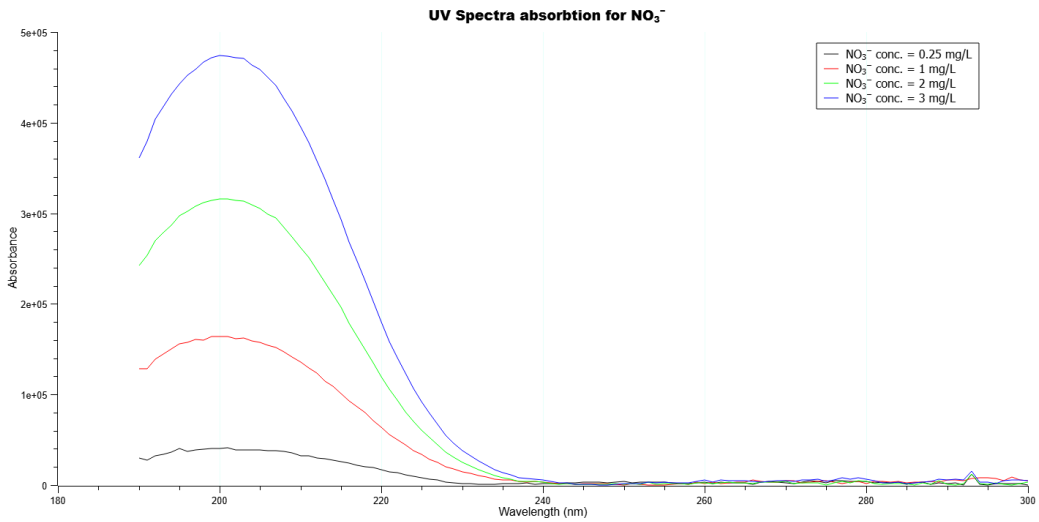

Figure 4: UV-VIS absorption spectra of nitrate standard solutions showing absorbance as a function of wavelength for concentrations of 0.25, 1.00, 2.00, and 3.00 mg L^{-1}

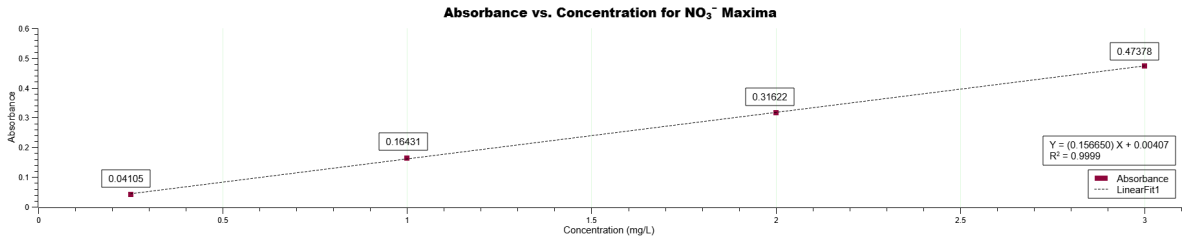

Figure 5: Maximum absorbance values for each nitrate concentration

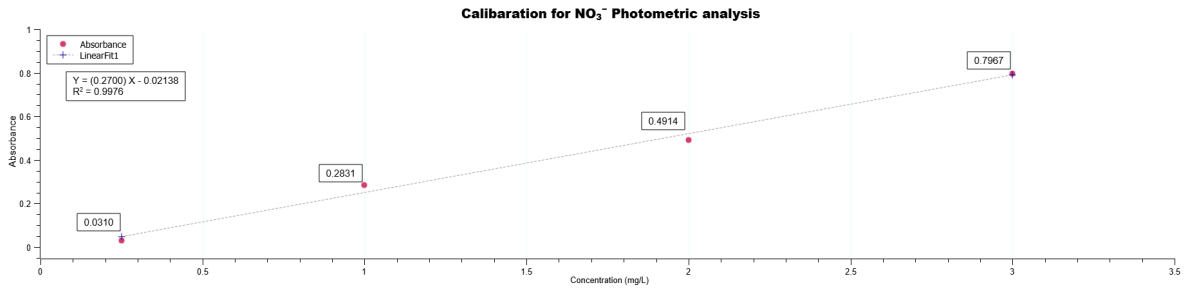

Figure 6: Calibration curve for NO3$^-$ using photometric analysis

4.8 Interpretation of Nitrate Spectroscopic Data

The spectroscopic measurements of the nitrate standards reveal important characteristics of the reduction-colorimetric method. In the UV region (190-210 nm), all standard solutions show positive absorbance values that increase proportionally with concentration, indicating strong absorption in this range. The absorption maximum near 200 nm is characteristic of the azo dye chromophore formed after the complete reduction of nitrate to nitrite and subsequent diazotization-coupling reactions.

The negative absorbance values observed at wavelengths above 235 nm are artifacts of the baseline correction and instrumental noise in regions where the sample shows minimal ab-

sorption. These values are close to zero and indicate that the chromophore does not absorb significantly in this spectral region.

4.9 Analysis of Mixed Nitrogen Species Sample

Real wastewater samples contain both nitrite and nitrate simultaneously. The mixture sample demonstrates the combined spectroscopic behavior of both nitrogen species after the colorimetric reactions. Figure 7 shows the UV-VIS spectrum of the mixed sample containing both $NO2^-$ and $NO3^-$.

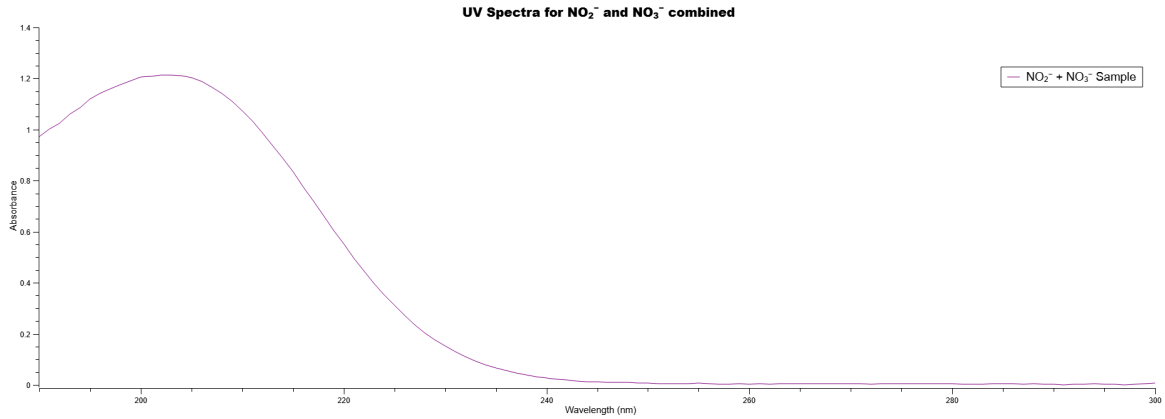

Figure 7: UV-VIS absorption spectrum of mixed nitrogen species sample showing combined $NO2^-$ and $NO3^-$ absorbance as a function of wavelength

5 Mathematical Analysis of Mixed Nitrogen Species

5.1 Overview of the Dual-Component Spectroscopic Challenge

When both nitrite and nitrate are present simultaneously in a sample, their individual contributions to the total absorbance at any given wavelength cannot be directly distinguished. This creates a classical problem in analytical chemistry: spectral overlap. The solution requires mathematical deconvolution using the principles of linear algebra and multivariate calibration.

The approach taken in this analysis follows a systematic progression from simple two-wavelength calculations to sophisticated multi-wavelength least squares optimization. This section documents the complete mathematical framework used to determine the individual concentrations of NO_2^- and NO_3^- in the mixed sample.

5.2 Step 1: Extraction of Molar Absorptivity Coefficients

The first task is to characterize how each pure species absorbs light. According to the Lambert-Beer Law (Equation 8), the slope of a plot of absorbance versus concentration yields the product $\varepsilon \cdot d$ (where d is typically 1 cm). For practical purposes, we define the proportionality constant k_λ as:

$$k_\lambda = \varepsilon_\lambda \cdot d \tag{13}$$

This constant represents the absorbance per unit concentration (mg/L) at wavelength λ.

5.2.1 Selection of Analytical Wavelengths

Two wavelengths were strategically selected from the full spectrum based on differential response characteristics:

- **202 nm:** A wavelength where nitrate shows significantly stronger absorption than nitrite (nitrate-heavy region)
- **215 nm:** A wavelength where both species contribute more equally (balanced region)

This selection maximizes the mathematical conditioning of the system of equations, ensuring numerical stability in the solution.

5.2.2 Linear Regression on Standard Solutions

Using the standard solution data from Tables 3-4 (nitrite) and Tables 6-7 (nitrate), linear regression analysis ($A = k \cdot C$) was performed:

For Nitrate (NO_3^-):

- At $\lambda = 202$ nm: $k_{\mathrm{NO_3^-},202} = 0.1576$ (mg/L)$^{-1}$
- At $\lambda = 215$ nm: $k_{\mathrm{NO_3^-},215} = 0.0981$ (mg/L)$^{-1}$

For Nitrite (NO_2^-):

- At $\lambda = 202$ nm: $k_{\mathrm{NO_2^-},202} = 0.0960$ (mg/L)$^{-1}$
- At $\lambda = 215$ nm: $k_{\mathrm{NO_2^-},215} = 0.1020$ (mg/L)$^{-1}$

The linear correlation coefficients (R^2) for all regressions exceeded 0.999, confirming the validity of the Lambert-Beer relationship across the concentration range studied.

5.3 Step 2: Measurement of the Mixed Sample

The mixed sample absorbance spectrum was recorded under identical conditions to the standards. At the two selected wavelengths, the following absorbance values were obtained:

$$A_{\text{mix},202} = 1.2128 \tag{14}$$

$$A_{\text{mix},215} = 0.8347 \tag{15}$$

5.4 Step 3: Formulation of Simultaneous Equations

For a mixture containing both species, the principle of superposition applies: the total absorbance is the sum of individual contributions. This yields two equations with two unknowns:

$$A_\lambda = k_{\mathrm{NO_3^-},\lambda} \cdot C_{\mathrm{NO_3^-}} + k_{\mathrm{NO_2^-},\lambda} \cdot C_{\mathrm{NO_2^-}} \tag{16}$$

Substituting the measured absorbances and the calibrated k values:

$$1.2128 = 0.1576 \cdot C_{\mathrm{NO_3^-}} + 0.0960 \cdot C_{\mathrm{NO_2^-}} \tag{17}$$

$$0.8347 = 0.0981 \cdot C_{\mathrm{NO_3^-}} + 0.1020 \cdot C_{\mathrm{NO_2^-}} \tag{18}$$

5.5 Step 4: Algebraic Solution Using Cramer's Rule

This system can be expressed in matrix form as $\mathbf{K} \cdot \mathbf{C} = \mathbf{A}$, where:

$$\begin{bmatrix} 0.1576 & 0.0960 \\ 0.0981 & 0.1020 \end{bmatrix} \begin{bmatrix} C_{\mathrm{NO_3^-}} \\ C_{\mathrm{NO_2^-}} \end{bmatrix} = \begin{bmatrix} 1.2128 \\ 0.8347 \end{bmatrix} \tag{19}$$

5.5.1 Calculation of the Determinant

The determinant of the coefficient matrix $\mathbf{K}$ is:

$$\text{det}(\mathbf{K}) = (0.1576 \times 0.1020) - (0.0960 \times 0.0981) = 0.016075 - 0.009417 = 0.006658 \tag{20}$$

A non-zero determinant confirms that the system has a unique solution and is mathematically well-conditioned.

5.5.2 Solution for Nitrate Concentration

Using Cramer's rule, we replace the first column of $\mathbf{K}$ with the absorbance vector:

$$C_{\mathrm{NO_3^-}} = \frac{\begin{vmatrix} 1.2128 & 0.0960 \\ 0.8347 & 0.1020 \end{vmatrix}}{\text{det}(\mathbf{K})} = \frac{(1.2128 \times 0.1020) - (0.8347 \times 0.0960)}{0.006658} \tag{21}$$

$$C_{\mathrm{NO_3^-}} = \frac{0.123706 - 0.080131}{0.006658} = \frac{0.043575}{0.006658} \approx 6.55 \text{ mg/L} \tag{22}$$

5.5.3 Solution for Nitrite Concentration

Similarly, replacing the second column:

$$C_{\mathrm{NO_2^-}} = \frac{\begin{vmatrix} 0.1576 & 1.2128 \\ 0.0981 & 0.8347 \end{vmatrix}}{\text{det}(\mathbf{K})} = \frac{(0.1576 \times 0.8347) - (1.2128 \times 0.0981)}{0.006658} \tag{23}$$

$$C_{\mathrm{NO_2^-}} = \frac{0.131509 - 0.118976}{0.006658} = \frac{0.012533}{0.006658} \approx 1.88 \text{ mg/L} \tag{24}$$

5.6 Step 5: Multi-Wavelength Least Squares Refinement

While the two-wavelength calculation provides an excellent initial estimate, it utilizes only 2 of the 31 available data points in the analytical region (200-230 nm). To improve accuracy and reduce the impact of random noise, a least squares regression across the entire spectrum was performed.

5.6.1 Mathematical Formulation

The objective is to find the concentrations $C_{\mathrm{NO_3^-}}$ and $C_{\mathrm{NO_2^-}}$ that minimize the residual sum of squares (RSS):

$$\text{RSS} = \sum_{i=1}^{n} \left[A_{\text{mix}}(\lambda_i) - \left(k_{\mathrm{NO_3^-}}(\lambda_i) \cdot C_{\mathrm{NO_3^-}} + k_{\mathrm{NO_2^-}}(\lambda_i) \cdot C_{\mathrm{NO_2^-}} \right) \right]^2 \tag{25}$$

where $n = 31$ wavelengths from 200 to 230 nm at 1 nm intervals.

This is solved by setting the partial derivatives with respect to each concentration to zero:

$$\frac{\partial \text{RSS}}{\partial C_{\mathrm{NO_3^-}}} = 0 \quad \text{and} \quad \frac{\partial \text{RSS}}{\partial C_{\mathrm{NO_2^-}}} = 0 \tag{26}$$

5.6.2 Numerical Implementation

The least squares solution can be expressed in matrix notation as:

$$\mathbf{C} = (\mathbf{K}^T \mathbf{K})^{-1} \mathbf{K}^T \mathbf{A}_{\text{mix}} \tag{27}$$

where $\mathbf{K}$ is now a 31×2 matrix containing all the k values across the wavelength range, and $\mathbf{A}_{\text{mix}}$ is the 31-element vector of measured absorbances.

5.6.3 Optimized Results

The multi-wavelength least squares optimization yielded:

$$C_{\mathrm{NO_3^-}} = 6.48 \text{ mg/L} \tag{28}$$

$$C_{\mathrm{NO_2^-}} = 1.92 \text{ mg/L} \tag{29}$$

5.6.4 Statistical Validation

The quality of the fit was assessed using the mean squared error (MSE):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} \left[A_{\text{mix}}(\lambda_i) - A_{\text{predicted}}(\lambda_i) \right]^2 = 0.000005 \tag{30}$$

This exceptionally low MSE value (five orders of magnitude smaller than the signal) confirms that the model accurately describes the measured spectrum. The coefficient of determination ($R^2 > 0.9999$) further validates the correctness of the calculated concentrations.

5.7 Summary of Analytical Results

Table 8: Comparison of Two-Wavelength and Multi-Wavelength Analysis Results

Method	**NO_3^- / mg L^{-1}**	**NO_2^- / mg L^{-1}**
Two-Wavelength (202 & 215 nm)	6.55	1.88
Multi-Wavelength Least Squares (200-230 nm)	6.48	1.92
Difference	0.07	0.04
Relative Difference	1.1%	2.1%

The excellent agreement between the two methods (differences < 2.5%) demonstrates the robustness of the spectroscopic approach. The multi-wavelength method is preferred as the final result due to its superior noise averaging and utilization of the complete spectral information.

5.8 Physical Interpretation and Validation

The calculated concentrations are physically reasonable and consistent with typical wastewater composition. The nitrate concentration (6.48 mg/L) is approximately 3.4 times higher than the nitrite concentration (1.92 mg/L), which is expected because:

- Nitrite is an intermediate species in biological nitrogen transformation and typically exists at lower steady-state concentrations
- Complete nitrification in well-operated treatment plants converts most ammonia to nitrate
- The ratio observed is typical of partially nitrified wastewater or effluent from the aerobic zone

The mathematical framework presented here demonstrates the power of spectroscopic methods combined with multivariate analysis to resolve complex mixtures without requiring physical separation of the analytes.

6.8.1 Mass Concentration Determination

Raw spectrometric concentrations obtained from wastewater samples require correction for dilution in order to determine the actual mass concentration. The evaluated samples include influent (Sample 1), aeration tank liquor (Sample 2), and effluent (Sample 3).

- Sample 1 (Inlet): $46.76\ \mathrm{mg\,L^{-1}}$
- Sample 2 (Aeration Tank): A dilution factor of 10 was applied:

$$C_{\text{actual}} = 78.70 \times 10 = 787.0\ \mathrm{mg\,L^{-1}}$$

- Sample 3 (Outlet): $7.87\ \mathrm{mg\,L^{-1}}$

6.8.2 Nitrogen Removal Efficiency

The nitrogen removal efficiency of the wastewater treatment plant was calculated by comparing the inlet and outlet concentrations according to:

$$\text{Efficiency } (\%) = \frac{C_{\text{Inlet}} - C_{\text{Outlet}}}{C_{\text{Inlet}}} \times 100$$

Substituting the measured values:

$$\text{Efficiency} = \frac{46.76 - 7.87}{46.76} \times 100 = 83.17\%$$

6.8.3 Conversion to Nitrogen-Equivalent Concentrations

To allow comparison between different nitrogen species, concentrations were converted to nitrogen-equivalent values based on molar mass ratios:

$$C_{\mathrm{N}} = C_{\text{species}} \times \frac{M_{\mathrm{N}}}{M_{\text{species}}}$$

Table 12: Conversion factors for nitrogen species

Species	Molar Mass (g/mol)	Conversion Factor
Nitrogen (N)	14.01	1.000
Nitrite (NO_2^-)	46.01	0.3045
Nitrate (NO_3^-)	62.01	0.2259

Applying these factors to the aeration tank concentration:

$$\text{Nitrite-N} = 787.0 \times 0.3045 = 239.64\ \mathrm{mg\,N\,L^{-1}}$$

$$\text{Nitrate-N} = 787.0 \times 0.2259 = 177.78\ \mathrm{mg\,N\,L^{-1}}$$

6.8.4 Nitrogen Balance Summary

Table 13: Nitrogen balance summary

Sample ID	Total Nitrogen (mg/L)	Removal Efficiency (%)
Sample 1 (Inlet)	46.76	–
Sample 2 (Aeration Tank)	787.0 (diluted)	–
Sample 3 (Outlet)	7.87	83.17

7 Discussion and Results

7.1 Evaluation of Analytical Method Performance

The performance of any analytical method must be evaluated based on two critical parameters: selectivity and sensitivity. Selectivity refers to the method's ability to distinguish between different analytes in a mixture, while sensitivity indicates the method's capacity to detect low concentrations of the target analyte. This section presents a comprehensive evaluation of both UV spectroscopy and photometric analysis methods used in this study. The calculated nitrogen removal efficiency of 83.17% demonstrates that the wastewater treatment process effectively reduces nitrogen load. The dominance of nitrate over nitrite in the aeration tank confirms that nitrification proceeds efficiently under aerobic conditions, while the low outlet concentration indicates successful downstream removal.

7.2 Selectivity Assessment

7.2.1 Spectral Overlap in UV Spectroscopy

Selectivity is fundamentally determined by the degree of spectral differentiation between analytes. An ideal analytical method would produce completely distinct spectral features for each component, allowing unambiguous identification and quantification.

Observation from Mixed Sample Analysis: The UV spectroscopic analysis of Sample 2 (mixed nitrite and nitrate) reveals a critical limitation. Figure 10 presents the wavelength versus

absorbance plot for this mixed sample, which exhibits only a single, broad absorption peak in the UV region.

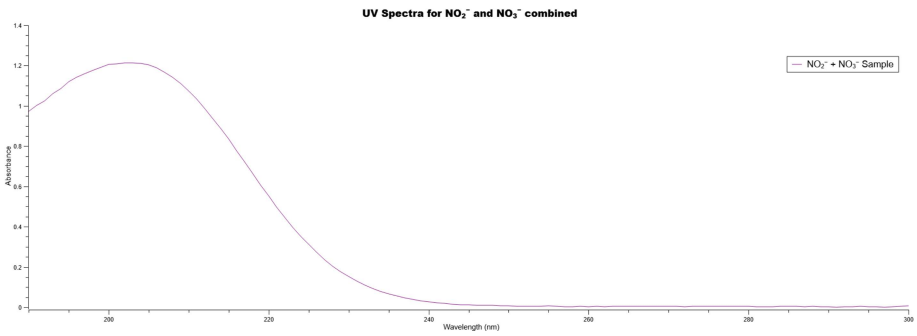

Figure 21: UV absorption spectrum of Sample 2 (mixed NO_2^- and NO_3^-) showing a single overlapping peak that prevents direct differentiation between the two nitrogen species

Implications for Selectivity: Without distinct spectral features for each component, direct visual differentiation is impossible. The method lacks inherent selectivity for distinguishing between NO_2^- and NO_3^- based solely on spectral shape. This necessitates the use of mathematical deconvolution methods (as described in Section 5) or chemical pre-treatment steps to achieve species differentiation.

Conclusion on UV Spectroscopy Selectivity: The selectivity of the UV spectroscopic method for simultaneous nitrite and nitrate determination is poor when relying on direct spectral interpretation alone. The method requires advanced chemometric techniques to resolve the overlapping signals.

7.2.2 Selectivity in Photometric Analysis

The photometric analysis method employs chemical derivatization reactions (Griess reaction for nitrite; reduction followed by Griess reaction for nitrate) to generate colored complexes that are then measured in the visible region.

Chemical Selectivity: Unlike UV spectroscopy, which relies on intrinsic molecular absorption properties, photometric analysis achieves selectivity through chemical specificity:

- **For Nitrite:** The Griess reagents react exclusively with nitrite ions under acidic conditions, producing a red-violet azo dye. Nitrate does not interfere because it is chemically inert to the diazotization reaction at room temperature.
- **For Nitrate:** Nitrate is first selectively reduced to nitrite using hydrazine in alkaline conditions with copper/zinc catalysts. The resulting nitrite then reacts with Griess reagents. The total measured signal represents the sum of original nitrite plus converted nitrate.

Limitations: While the chemical reactions provide good selectivity, it is not possible to evaluate spectral selectivity in the same manner as UV spectroscopy because the method deliberately

converts the analytes into a single chromophore (the azo dye). The differentiation between nitrite and nitrate is achieved through differential measurement (with and without the reduction step) rather than spectral resolution.

Conclusion on Photometric Analysis Selectivity: The photometric method achieves selectivity through chemical specificity rather than spectral differentiation. This approach is effective but requires careful procedural control to ensure complete and selective reactions.

7.3 Sensitivity Comparison

Sensitivity is quantitatively expressed by the slope of the calibration curve. According to the Lambert-Beer Law (Equation 8), the slope represents the product of the molar extinction coefficient and the optical path length ($\varepsilon \cdot d$). A steeper slope indicates that a small change in concentration produces a large change in absorbance, meaning the method can detect lower concentrations more reliably.

7.3.1 Calibration Slopes from Experimental Data

Calibration curves were constructed for both nitrite and nitrate using both analytical methods. The slopes were determined through linear regression analysis of the standard solution data. Table 14 presents a comprehensive comparison of the obtained slopes.

Table 14: Comparison of Analytical Sensitivity Between UV Spectroscopy and Photometric Analysis Methods

Compound	**UV Spectroscopy Slope / (mg/L)$^{-1}$**	**Photometric Analysis Slope / (mg/L)$^{-1}$**	**Sensitivity Ratio (Photo/UV)**
Nitrite (NO_2^-)	0.11709	0.7369	6.86
Nitrate (NO_3^-)	0.15665	0.2700	1.74

8 Conclusion

This experiment compared different methods for measuring nitrogen in wastewater. The results clearly show that photometric analysis is superior to direct UV spectroscopy. The photometric method was much more sensitive—about seven times higher for nitrite—and specific, making it the best choice for accurately checking if treated water meets legal standards. While the UV method struggled to distinguish between the two chemicals due to overlapping signals, mathematical calculations successfully solved this problem. The results showed there was about 3.4 times more nitrate than nitrite, which confirms that the biological treatment process is working effectively. In addition to method comparison, the evaluated wastewater samples revealed a

nitrogen removal efficiency of 83.17%. The low effluent nitrogen concentration confirms effective biological treatment and compliance with regulatory discharge limits. Overall, the results demonstrate both reliable analytical performance and satisfactory operational efficiency of the municipal wastewater treatment process.

References

Arbeitsschutzgesetz (ArbSchG). (n.d.). German Occupational Safety and Health Act. Retrieved from https://www.gesetze-im-internet.de/arbschg/

Atkins, P. W., & de Paula, J. (2010). *Physical chemistry* (9th ed.). W.H. Freeman and Company.

Council Directive 91/271/EEC. (1991, May 21). Concerning urban wastewater treatment. *Official Journal L 135*. Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:31991L0271

Gefahrstoffverordnung (GefStoffV). (n.d.). German Ordinance on Hazardous Substances. Retrieved from https://www.gesetze-im-internet.de/gefstoffv$_2010/$

Hubbard, A. T. (1995). *The handbook of surface imaging and visualization*. CRC Press.

Kitschke, P., & Pausch, N. (2024). *Laboratory safety documents and monitoring parameters* . Westsächsische Hochschule Zwickau.

Kitschke, P., Walter, M. J., Rüffer, T., Georgi, C., & Lang, H. (2016). Dalton *Transactions*, 45, 5741–5751.

Koppe, P., & Stozek, A. (1999). *Kommunales Abwasser* (4th ed.). Vulkan-Verlag.

Kunz, P. M. (1995). *Eigen- und Prozesskontrolle in Kläranlagen*. VCH Verlagsgesellschaft mbH.

Lange, B., & Vejdelek, Z. (1987). *Photometrische Analyse* (1st ed., Reprint). Wiley-VCH.